

LAVOIR: Teaching a Single-Pass Decision Encoder When and What to Ask with Amortized Value of Information

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Abstract

“System One” decision models such as TypeSafe’s Jev and its open counterpart Laya answer typed questions about a text in a single forward pass with calibrated probabilities, but they cannot ask for missing information: when a first message does not say what separates two departments, they guess. We present LAVOIR (**L**aya with **V**alue-**O**f-**I**nformation **R**outing), which places the candidate pieces of missing information (*slots*) in the input next to the answer options, so that one forward pass returns both the decision distribution and, for every slot, the expected gain in the probability of the correct decision if the user were asked about it. VOI targets need no human labels: gold decisions come from schema rules, an LLM only verbalizes messages and answers, a model from another family checks every text, and pairing each message with several profiles makes regression on realized gains estimate the expected gain. A Gini-impurity cap bounds the predicted value by what a calibrated model can still gain. In a controlled study, decisions on seen schemas are statistically indistinguishable from the Bayes ceiling. The final model’s question policy matches a greedy oracle VOI policy on seen schemas (AUC 0.799 vs. 0.797), and with at most 0.5 questions per conversation it is 14.1 points more accurate than never asking. On real ABCD conversations, one real exchange raises accuracy by 8.3 points where LAVOIR asks and leaves it unchanged where it does not; on SGD the cap lowers the asking rate from 93% to 8.6%. On Laya’s twelve benchmarks LAVOIR is above Laya’s reported scores on seven, and it answers a question in 31 ms (median, GH200).

1 Introduction

Many production decisions over text are small, typed and frequent: which team should handle a ticket, whether an e-mail is phishing, which tool to call. LLMs can make them, but they generate text token by token and do not expose calibrated probabilities. TypeSafe introduced Jev (Almeida, 2026) as the first *System One* model, named after the fast mode of thinking in Kahneman (2011): it returns typed decisions with probabilities from one parallel query, and its weights are not public. Laya (Convai Innovations, 2026), released three days later as an open model with a Jev-compatible interface, places

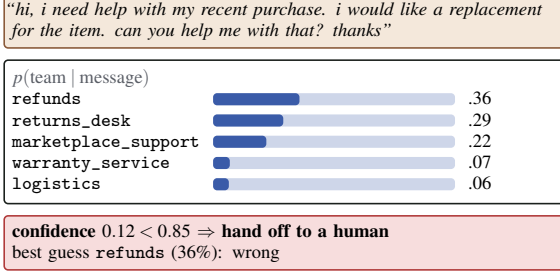
every answer option behind its own [MASK] marker in the input of a bidirectional encoder (Warner et al., 2024) and scores all options in one pass; the question schema is given at request time, so a new decision needs no retraining.

These models must decide from the text they are given, and real first messages are often underspecified. “*My information was shared without permission, what can I do?*” may belong to the data-protection or to the legal team, depending on what was shared and what the customer wants. A human agent asks one question; a single-pass model guesses or escalates every uncertain case (Figure 1).

LAVOIR adds a second marker block for slots after the answer options. In the same forward pass that produces the decision distribution, a small *VOI head* reads each slot marker and predicts how much asking about that slot would raise the probability of the correct decision; the model asks about the best slot if its value exceeds a question cost and otherwise decides or hands off (Figure 2). The question *content* is chosen by the model; its *wording* is a template attached to the slot. The training signal needs no human labels: schema rules map complete user profiles to gold units, an LLM only verbalizes partial profiles, and because every message is paired with several profiles that differ in their hidden values, least-squares regression on the realized gain $p_k(y^*) - p_0(y^*)$ estimates the expected gain, an amortized value of information (Howard, 1966; Rao and Daumé III, 2018). Our contributions are:

- **Amortized VOI for typed decisions:** a slot block, a zero-initialized segment embedding (with no slots the model reproduces Laya’s logits bit for bit), a VOI head, and a Gini cap that bounds predicted value by the maximum expected gain of a calibrated model.
- **Label-free VOI targets** from rule-defined gold with an exact posterior, an answer simulator, and two-way cross-family checking, together with the schema design lessons that made generation work.
- **An evaluation against exact references:** the decisions match the Bayes ceiling and the question policy matches an oracle policy; on real conversations the questions go where answers help.
- **Findings about Laya’s recipe:** its policy-gradient term brings no gain and slows the VOI head, and one temperature per question type miscalibrates heterogeneous

Laya: decide or hand off



LAVOIR: ask, then decide

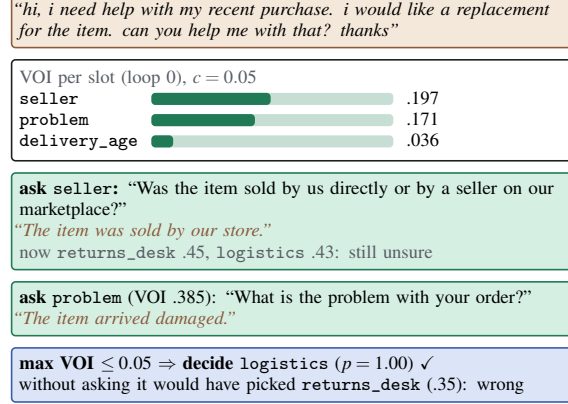


Figure 1: The same underspecified first message (a case of the `ecommerce_returns` schema) given to Laya and to LAVOIR. Laya’s confidence (one minus the normalized entropy, as Laya computes it) stays at 0.12, so its recommended usage (hand off below 0.85) sends the conversation to a human, and its best guess is wrong anyway; Laya never saw this schema in training. LAVOIR asks about the seller and then about the problem, skips the three slots whose answers would not change the decision, and routes to the correct team.

mixtures.

2 Related Work

System One models and decision encoders. Jev (Almeida, 2026) returns typed answers (categories, scores, a choice among up to 255 options) with probabilities, decodes in parallel and is trained with what TypeSafe calls Reinforcement Learning for Calibrated Decisions (RLCD); it is API-only, with 236–276 ms median latency in independent measurements (AbdelStark, 2026; nibzard, 2026). Laya (Convai Innovations, 2026) implements this interface on ModernBERT-large (Warner et al., 2024) or mmBERT (Marone et al., 2025), trains with strictly proper scoring rules (Gneiting and Raftery, 2007) plus a group-baseline policy-gradient term (Williams, 1992; Shao et al., 2024), and calibrates with per-type temperatures (Guo et al., 2017). GLiNER and GLiClass (Zaratiana et al., 2024; Stepanov et al., 2025) encode labels jointly with the text, and SCX Router (Stepanov et al., 2026) scores label tokens against a decoder cache for model routing (Ong et al., 2024). None of these models can ask for missing information.

Clarifying questions. Rao and Daumé III (2018) rank clarification questions by the expected value of perfect information, generating candidate answers. Qulac and ClariQ (Aliannejadi et al., 2019, 2021) study clarification in search, Yu et al. (2020) ask binary questions chosen by information gain, and LLM-based work decides when to clarify (Kuhn et al., 2022; Zhang and Choi, 2023), trains models to ask (Andukuri et al., 2024), or simulates answers to plan questions (Hu et al., 2024). LAVOIR instead predicts the value of every candidate question in one encoder pass, without generating or simulating answers at inference.

Value of information, deferral and conformal prediction.

VOI (Howard, 1966) and expected information (Lindley, 1956) are the classical criteria for choosing what to observe; deep adaptive design amortizes sequential design into a policy network (Foster et al., 2021), and POMDP dialogue managers trade off asking and acting with simulated users (Williams and Young, 2007; Schatzmann et al., 2007). Selective classification (Geifman and El-Yaniv, 2017) and learning to defer (Mozannar and Sontag, 2020) decide whether to answer at all; KnowNo (Ren et al., 2023) asks for help when a conformal set (Vovk et al., 2005; Angelopoulos and Bates, 2021) has more than one option; our baseline B5 uses the same trigger with an uncalibrated prediction set. Because LLM-generated data carries exploitable artifacts (Li et al., 2023; Gururangan et al., 2018; Geirhos et al., 2020), we keep labels out of the LLM and check texts with a model from another family.

3 Problem Formulation

We follow Laya’s interface: a *state* s (a message or JSON object) and a *question* of type *choice*, *score* or *noul* (yes/no) with options $o_1, \dots, o_m$; the model returns $p(\cdot \mid s, q)$.

Schemas and the exact posterior. A *schema* has a set of units $\mathcal{U}$ (the options), decisive and non-decisive slots, a finite value set V_k for every decisive slot, a conditional prior $P(v_k \mid v_{\pi(k)})$ with at most one parent slot, and an ordered rule list g (the last rule is *else*) that maps decisive-slot values to a unit. A *profile* θ assigns every slot; its gold unit is $y^* = g(\theta)$. What the model has seen about slot k is an evidence set $E_k \subseteq V_k$: $\{\theta_k\}$ for a stated slot or a clean answer, $\{\theta_k, a\}$ for a *partial* answer (“it might be a or maybe b ”), V_k for “I don’t know”; an *over-informative* answer also reveals one unasked slot. With a uniform likelihood inside each evidence set,

$$P(y \mid E) \propto \sum_{\theta} P(\theta) \mathbb{I}[g(\theta) = y] \prod_k \mathbb{I}[\theta_k \in E_k], \quad (1)$$

computed exactly by enumeration. The *Bayes ceiling* of a test set is the mean of $\max_y P(y | E)$; no text-only model can exceed it in expectation.

Value of information. With $A_k(\cdot | \theta)$ the simulator’s answer distribution for slot k , the *oracle VOI* is the expected increase in the probability of the correct unit:

$$\text{VOI}^*(k | E) = \mathbb{E}_{\theta \sim P(\cdot | E)} \mathbb{E}_{a \sim A_k(\cdot | \theta)} [P(g(\theta) | E \oplus a)] - \sum_y P(y | E)^2. \quad (2)$$

Proposition 1 (Gini bound). $\text{VOI}^*(k | E) \leq 1 - \sum_y P(y | E)^2 = G(P(\cdot | E))$, the *Gini impurity* (Breiman *et al.*, 1984) of the posterior.

Proof. The first term of Eq. (2) is an expected probability, hence at most 1. $\square$

For a calibrated model $p \approx P(\cdot | E)$, the value of any question therefore vanishes as the model becomes certain; §7.2 shows that a learned head does not discover this on its own out of distribution.

Question policy. Given p and predicted values $\hat{v}_k$ for the slots not yet asked, LAVOIR asks about $k^* = \arg \max_k \hat{v}_k$ if $\hat{v}_{k^*} > c$ and fewer than Q questions have been asked, appends the question and answer to the state and re-encodes; otherwise it hands off if $1 - \max_y p_y > c_h$ and decides $\arg \max_y p_y$ if not (Figure 2). A slot is never asked twice.

4 Model

Input. Laya’s sequence has [CLS], the question type and instruction, one [MASK] per option, and the state. We append an optional slot block introduced by the plain-text header “missing information:” (Figure 3). Option and slot orders are re-shuffled every time an example is read, and the state is always a list of turns, so the target snapshot and the deployed model see the same format.

Architecture. We keep Laya’s decision head on ModernBERT-large (a question-type embedding, two transformer layers and an option scorer; Convai Innovations, 2026) and add three components. A *segment embedding* of the block id is initialized to zero, so with an empty slot block the logits are bit-identical to Laya’s (`torch.equal`, real weights, fp32). The *VOI head* computes $h_k = \text{MLP}([z_k; f(p)])$ (LayerNorm over $d + 4$ inputs, 256 GELU units, scalar output; 0.27M parameters), where z_k is the head output at slot marker k and $f(p)$ holds the largest probability, the top-two margin, the entropy divided by $\log m$, and $m/255$ (the features of Laya’s act head), computed on the detached distribution. The *Gini cap* gives

$$\hat{v}_k = G(\text{softmax}(z/T)) \cdot \sigma(h_k), \quad G(p) = 1 - \sum_y p_y^2, \quad (3)$$

with G detached, so a confident model predicts a structurally small value. Uncapped variants output h_k directly.

Losses. Laya’s decision loss is a soft cross-entropy plus a policy-gradient term that scores noisy logits with a proper reward (Gneiting and Raftery, 2007; Epstein, 1969) and applies REINFORCE with a group-mean baseline; we expose its weight w_{RL} (1 is Laya’s recipe). The VOI target of slot k is

$$t_k = \hat{p}_k(y^*) - \hat{p}_0(y^*), \quad (4)$$

where $\hat{p}_0$ is a frozen, temperature-scaled snapshot on the example and $\hat{p}_k$ the same after a sampled answer to slot k (a *probe*) is appended. We minimize the MSE between $\hat{v}_k$ and t_k , both divided by the target standard deviation, with weight $\lambda = 1$. The target uses the gold unit, not the model’s own risk: a self-referential target such as the drop in $1 - \max_y p_y$ rewards any answer that makes the model confident, including a misleading one. It is also bounded in $[-1, 1]$, unlike a log-loss difference. Because each message has several profiles, the regression estimates the snapshot’s version of Eq. (2).

Training. A *warm* phase trains the decision model on all examples (500 frozen-encoder steps, then four epochs; learning rates 2.5×10^{-5} encoder and 10^{-4} head, cosine, bf16, global batch 64). Temperatures are fitted with L-BFGS (Liu and Nocedal, 1989) on a 5% held-out slice. The warm snapshot computes targets, two *joint* epochs train $\mathcal{L}_{\text{dec}} + \lambda \mathcal{L}_{\text{VOI}}$ with encoder learning rate 5×10^{-6} , and targets are recomputed from that snapshot for one more joint epoch, so the head regresses on values of a decision model that has stopped moving.

5 Data

Schemas. We wrote twelve customer-service schemas, eight for training and four held out for zero-shot evaluation, each with five or six units, three or four decisive slots and one non-decisive slot (Appendix A). Before generating any text, automatic gates checked reachability, unit masses, the share of units that need two or more slots, and the mean maximum posterior and VOI share of a dry run. Three pilots and one discarded full run showed that data quality is governed by the *semantic independence* of slots, not by rule complexity: overlapping slots (“an unrecognized transaction” and “did you authorize it”) leak each other; a *not applicable* value is indistinguishable from “no” in text; default-like values (“technical help”) are leaked by any vague message; non-decisive dates or amounts imply decisive facts, so the final non-decisive slot is the customer’s name; and the generator must be forbidden to invent specifics when the topic is hidden.

Cases and answers. A *case* is a partial profile (the slots stated in the first message) with two to four complete profiles consistent with it, which differ in their hidden values and, typically, in their gold unit; without this pairing the regression would learn a deterministic mapping rather than an expectation. Each profile yields a message-only example, with probability 0.7 one with a question–answer pair, and with probability

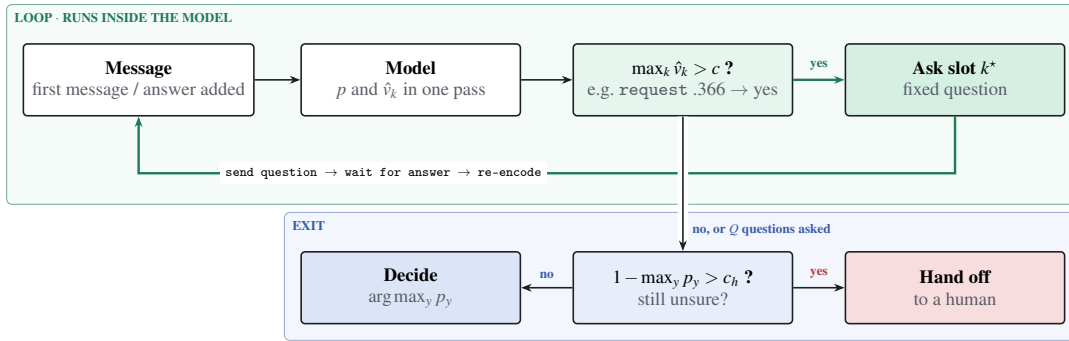


Figure 2: The asking loop of §3. Every forward pass returns the decision distribution p and a value $\hat{v}_k$ for each slot not yet asked. While the best slot’s value exceeds the cost c and the budget Q is not used up, its fixed question is sent and the answer is appended to the conversation, which is re-encoded. Otherwise the model decides, or hands the case to a human if it is still unsure. In the example of Figure 1 the loop runs twice before exiting.

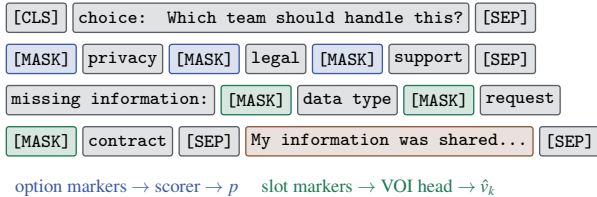


Figure 3: Input sequence. The second marker block lists the slots that could be asked; option markers (blue) feed the option scorer and slot markers (green) feed the VOI head in the same forward pass. The state (sand) is the conversation so far.

0.4 one with two, plus a probe for every slot. Answers are clean (0.675), partial (0.125), “I don’t know” (0.10) or over-informative (0.10); probes of already known slots are always clean.

Generation and two-way checking. Messages (ten styles, at most 120 words) and answers are written by Qwen3.6-35B-A3B (Yang et al., 2025; Qwen Team, 2026) with vLLM (Kwon et al., 2023). The prompt lists the facts to state plainly and the topics not to mention, without their values. Gemma-4-26B-A4B (Gemma Team, 2026) returns, for every decisive slot, a constrained guess and an evidence level; a text is re-generated (up to three times) if a hidden slot is guessed with explicit evidence (*leakage*) or a stated slot is not recovered (*fidelity*). The final run (two GH200 nodes, 12 min) produced 21,601 training examples (8×500 cases), 5,379 seen-schema test examples and 5,432 zero-shot test examples. First-attempt message rejection was 1–10% for seven training schemas and 22% for privacy, whose “type of data” and “request” slots overlap irreducibly; answer rejection was at most 2.8%, and partial answers were guessed correctly 46–62% of the time (expected 50%).

General single-turn data. To keep general ability we add single-turn data in Laya’s format (details in Appendix B): typed-decisions, CLINC150, MultiNLI, Yelp and SGD first turns (Convai Innovations, 2026; Larson et al., 2019; Williams

et al., 2018; Zhang et al., 2015; Rastogi et al., 2020) (19,000); tool choice from Nemotron post-training data (NVIDIA, 2025) (6,000, of which 5,447 train); the six sources Laya trained on with question-form yes/no items (Zhang et al., 2015; Clark et al., 2019; Metsis et al., 2006; Liu, 2024; Bajaj et al., 2016; Tobi-Bueck, 2025) (19,437), with all 4,447 benchmark texts removed; and 27 open sources (38,200, train splits only). *Controlled* models use the schema data, the first 19,000 and tool choice (46,048 examples); *broad* models use everything (103,685).

Real conversations. For out-of-distribution tests we use 918 ABCD test dialogues (Chen et al., 2021) with ten flows as units, the first customer block as state and the agent’s first reply plus the customer’s second block as the real exchange, and 3,812 SGD first user turns (Rastogi et al., 2020) from 20 services (2,766 from services unseen in training) with the service’s arguments as slots, which cannot change the intent.

6 Experimental Setup

Models. All models use ModernBERT-large and a single seed: *controlled* models with $w_{RL} \in \{0, 1\}$ (RL0/RL1) and either one temperature per question type (Laya’s procedure) or per source group; the *broad* model ($w_{RL} = 0$, per-source temperatures, uncapped); and the final LAVOIR, the broad warm model with the joint phases re-run with the Gini cap. Training took 28–48 min on 8–16 GH200 GPUs of the JUPITER booster. Laya and Jev numbers are those reported by Laya (Jev’s are third-party measurements); apart from Figure 1 we did not run Laya’s checkpoints.

Policies. All policies share the decision head and allow at most two questions: **B2** never asks; **B3** asks about a random unresolved slot when $\max_y p_y < \tau$; **B5** asks about the highest-VOI slot when the prediction set, the smallest set of units whose probabilities sum to at least $1 - \alpha$, contains more than one unit, with $\alpha \in \{0.5, 0.3, 0.2, 0.1, 0.05, 0.02, 0.01\}$ swept like the other thresholds (the set is not calibrated on held-out

Table 1: Controlled study on seen and zero-shot (zs) schemas. *Gap*: accuracy minus Bayes ceiling (95% CI). ρ , top-1: predicted vs. oracle VOI. AUC: area under the accuracy–questions curve (0–2 questions). b0.5: best accuracy with ≤ 0.5 questions per conversation. Upper block: group temperatures; lower: one temperature per type (Laya).

T	Arm	Split	Single turn			ECE	KL	VOI vs. oracle		AUC					b0.5		
			Acc	Ceiling	Gap [95% CI]			ρ	top-1	B2	B3	B5	VOI	ORACLE	B2	B3	VOI
group	RL1	seen	.653	.659	−.006 [−.016, .004]	.015	.016	.821	.876	.612	.766	.791	.800	.803	.612	.669	.741
group	RL0	seen	.652	.659	−.006 [−.017, .004]	.013	.012	.831	.923	.611	.763	.791	.799	.800	.611	.660	.755
group	RL1	zs	.517	.667	−.150 [−.162, −.137]	.130	.993	.690	.558	.502	.590	.607	.629	.658	.502	.566	.612
group	RL0	zs	.512	.667	−.154 [−.166, −.142]	.099	.747	.730	.745	.492	.574	.595	.616	.633	.492	.542	.579
single	RL1	seen	.652	.659	−.007	.078	.063	.817	.873	.614	.753	.772	.803	.802	.614	.614	.733
single	RL0	seen	.648	.659	−.010	.064	.043	.773	.934	.608	.751	.772	.797	.798	.608	.608	.762
single	RL1	zs	.514	.667	−.152	.071	.492	.682	.554	.499	.581	.580	.622	.653	.499	.499	.609
single	RL0	zs	.522	.667	−.145	.070	.481	.702	.649	.498	.578	.576	.624	.647	.498	.498	.549

data, so it carries no coverage guarantee); **VOI** is our policy; **ORACLE** asks about the slot with the largest exact VOI*. ORACLE is greedy (it looks one question ahead) and, like every policy, decides with the model’s own decision head, so it is a strong reference rather than an upper bound, and its value differs between models. For every message-only test example we run one dialogue per profile, answering with the profile’s probe, and sweep c , τ and c_h .

Metrics. Accuracy, Bayes ceiling and their gap with a bootstrap 95% interval (Efron and Tibshirani, 1993), ECE (Naeini et al., 2015) and KL to the posterior; Spearman ρ and top-1 agreement between $\hat{v}$ and VOI*; **AUC**, the normalized area under the upper envelope of accuracy vs. mean questions (0–2); **b0.5**, the best accuracy with at most 0.5 questions per conversation; and the share of *unnecessary* questions, those about a non-decisive slot or a slot whose value is already determined by the evidence, among all questions asked. AUC differences are given on the 0–1 scale, accuracy differences in percentage points.

7 Results

7.1 Controlled study

Decisions on seen schemas are Bayes-optimal. Accuracy is 0.006 below the Bayes ceiling with an interval that contains zero (Table 1); a model exploiting leaked hints would exceed the ceiling, and one under-using the text would fall clearly below it. Two schemas looked suspicious in argmax accuracy (insurance 0.649 vs. a ceiling of 0.610), but comparing the probability of the gold unit with the posterior under a case-clustered bootstrap shows no excess on any schema (Appendix C); the argmax excess comes from nearly tied posteriors. Per-schema leakage tests should therefore use probabilities, not argmax accuracy.

The question policy matches the oracle. With group temperatures the VOI policy reaches an AUC of 0.799–0.800 against 0.800–0.803 for ORACLE, with $\rho = 0.82$ –0.83 and 88–92% top-1 agreement. With at most 0.5 questions per conversation it is 12.9–14.4 points more accurate than never

Table 2: Broad model without and with the Gini cap (final LAVOIR). Question rates at $c = .02/.05/.1$. ABCD gains: accuracy change in points after the real first exchange in the cases where the policy (at $c = .02$) would and would not ask, with 95% paired bootstrap intervals over cases (10,000 resamples; 529/389 cases for LAVOIR, 844/74 uncapped). ORACLE is greedy and uses each model’s decision head, so it can lie below VOI.

	Uncapped	LAVOIR
SGD question rate	.93 / .87 / .69	.086 / .063 / .048
SGD AUROC(VOI $\rightarrow$ error)	.70	.84
SGD accuracy / ECE	.948 / .041	.942 / .044
ABCD question rate	.92 / .83 / .71	.58 / .49 / .41
ABCD AUROC(VOI $\rightarrow$ error)	.65	.71
ABCD accuracy / ECE	.653 / .222	.657 / .186
ABCD gain, asks	+2.7 [0.5, 5.0]	+8.3 [4.9, 11.7]
ABCD gain, does not ask	+2.7 [0.0, 6.8]	−0.3 [−1.5, 1.0]
Seen: AUC VOI / ORACLE	.795 / .797	.799 / .797
Seen: b0.5	.757	.751
Seen: ρ / top-1	.763 / .960	.851 / .937
Zero-shot: ρ / b0.5	.662 / .535	.689 / .581

asking (B2) and 7.2–9.5 points more accurate than asking at random when uncertain (B3). B5 uses the same VOI ranking but its AUC is 0.008–0.009 lower: deciding *whether* to ask from the VOI itself beats deciding it from the size of a prediction set. At $c = 0.02$ RL0 asks 1.09 questions per conversation for an accuracy of 0.889 with 1% unnecessary questions (oracle: 1.07 for 0.890); adding $c_h = 0.3$ hands off 18% of conversations and is 0.973 accurate on the rest.

Zero-shot schemas. On the four held-out schemas the VOI policy still beats every baseline (AUC 0.02–0.05 above B5; with group temperatures, b0.5 3.7–4.6 points above B3), but decisions are weak: 0.51–0.52 accuracy against a ceiling of 0.667. Trained on eight schemas, the model does not learn to read unseen rules from option descriptions.

7.2 Real conversations and the Gini cap

Without the cap the model asks too much. The broad model is 0.948 accurate on SGD and well calibrated (ECE 0.04), yet at $c = 0.02$ it would ask in 93% of SGD and 92% of ABCD conversations. Among the 3,526 SGD cases with con-

confidence ≥ 0.99 the median maximum VOI is 0.133, although by Proposition 1 the expected gain there is at most about 0.02. The head receives the decision distribution but has not learned the bound: out of distribution the slot representation dominates, and arguments that cannot change the intent get high value (amount 0.25, ride fare 0.18).

The cap fixes it structurally. Re-running only the joint phases with Eq. (3) leaves single-turn decisions essentially unchanged (calibration-slice accuracy $0.849 \rightarrow 0.847$, ABCD first-message accuracy $0.653 \rightarrow 0.657$) and drops the question rate on confident cases to zero (Table 2). On SGD at $c = 0.05$ the model asks in 40% of its errors and 4% of its correct decisions. On ABCD the uncapped model’s questions were untargeted: the real exchange helped equally where it would and would not ask (+2.7 each). With the cap the whole gain falls where it asks (+8.3, 95% CI [4.9, 11.7]) and none where it does not (−0.3 [−1.5, 1.0]); the difference between the groups is 8.6 [4.9, 12.2] points, against 0.0 [−4.7, 3.9] for the uncapped model, so the decision to ask now tracks the value of the answer. Over all conversations the capped model gains about 4.7 points from the exchange against 2.7 for the uncapped one: re-running the joint phases changed how the decision head reads a second turn, which our single-turn checks do not measure, so we compare the two groups within each model rather than the totals across models. On seen schemas the cap raises VOI quality (ρ 0.76 $\rightarrow$ 0.85) and keeps the AUC at ORACLE level; ORACLE is slightly below VOI here (0.797 vs. 0.799) because it is greedy and shares the decision head. With at most 0.5 questions the final model reaches 0.751, against 0.610 for never asking and 0.663 for random questions. Zero-shot b0.5 improves by 4.6 points. What remains is calibration: on ABCD the model is wrong in 19% of the cases where its confidence is at least 0.99, where the cap correctly stays silent but the confidence is wrong.

7.3 External benchmarks

We ran Laya’s benchmark builder and metrics unchanged on AG News (Zhang et al., 2015), DAIR Emotion (Saravia et al., 2018), Banking77 (Casanueva et al., 2020), support tickets (Tobi-Bueck, 2025), Enron spam (Metsis et al., 2006), phishing (Liu, 2024), ToxicChat (Lin et al., 2023), MS MARCO (Bajaj et al., 2016), a model-routing task over GSM8K and MBPP items (Cobbe et al., 2021; Austin et al., 2021), MASSIVE-en (FitzGerald et al., 2023) and the typed-decisions test split. LAVOIR is above Laya on seven of twelve sets, tied on spam and below on four (Table 3); Banking77, jailbreak and MASSIVE were never in our training data, and on typed-decisions it is also above Jev. Latencies were measured on different GPUs (ours GH200, Laya T4).

Yes/no collapse. The controlled model beat Laya on typed-decisions, Banking77, MASSIVE and jailbreak but collapsed on every yes/no benchmark (“false” on 398/400 spam cases, “true” on 399/400 RAG cases). All its yes/no training items were *assertions* (“This trace requires human review.”), while

Table 3: Accuracy on Laya’s benchmarks (seed 13, 400 cases per task). *Ctrl*: controlled RL0. Bold: LAVOIR above Laya. Laya: best released checkpoint; Laya and Jev as reported by Laya (Jev’s Banking77: 72 labels, $n = 100$). *Training split in our broad data and in Laya’s; †related train splits only. Latencies come from different hardware (ours GH200, Laya T4, Jev through its API) and are not directly comparable.

Set	Ctrl	LAVOIR	Laya	Jev
typed-decisions*	.785	.774	.766	.727
Banking77 (77)	.575	.533	.492	.870
MASSIVE-en (20 opt.)	.790	.805	.783	–
jailbreak (ToxicChat)	.888	.825	.762	–
model routing†	.283	.754	.659	–
toxicity (ToxicChat)	.525	.605	.530	–
RAG relevance*	.503	.665	.657	–
spam*	.480	.993	.993	–
phishing*	.625	.978	.993	–
AG News*	.765	.905	.953	.910
DAIR Emotion	.528	.575	.600	.480
support triage*	.285	.383	.522	–
latency, p50	27 ms	31 ms	33–40 ms	236 ms

benchmark instructions are *questions* over dictionaries with named fields. Question-form yes/no items and the open sources removed the collapse (spam $0.480 \rightarrow 0.993$) and lifted model routing from 0.283 to 0.754, at a small cost in VOI rank correlation on seen schemas.

7.4 Lessons about Laya’s recipe

The policy-gradient term does not help. In a miniature model, after the decision has converged the policy-gradient term keeps sending gradients to the shared parameters that are one to two orders of magnitude larger than the VOI gradient; with a fixed exploration scale its variance does not vanish at the optimum. With $w_{RL} = 0$ the VOI loss reached 10^{-4} in 200 steps, with $w_{RL} = 1$ it was still 0.06 after 1,000. At full scale the RL0 arm is ahead in every warm epoch, reaches the same decision accuracy (0.918 vs. 0.919), and learns the VOI head better (top-1 0.92 vs. 0.88 on seen and 0.75 vs. 0.56 on zero-shot schemas; Appendix C). Because Laya’s reward is a differentiable function of the reported distribution, its gradient can be taken directly; policy gradients would be needed only if the reward depended on which questions were asked.

One temperature per type is not enough. Laya fits one temperature per question type. On our mixture this gave $T = 2.25$ – 2.75 for choice because the general data has one-hot targets, while our schemas, whose targets are soft posteriors, were already calibrated at $T = 1$; the shared temperature made their KL ten times worse ($0.004 \rightarrow 0.04$ – 0.06). Per-group temperatures ($T \approx 1.02$ for our schemas) improve ECE on seen schemas five-fold, though not on zero-shot schemas (0.070 – $0.071 \rightarrow 0.099$ – 0.130 ; Table 1), yet the dialogue AUC barely moves: the VOI *ranking* is robust to temperature, and calibration matters mainly for hand-off and, through the cap, for staying silent.

8 Discussion and Conclusion

LAVOIR turns a single-pass decision encoder into one that knows when a question is worth asking and which one to ask, with the value of every candidate question predicted in the same forward pass as the decision. Rule-defined gold, several profiles per message, two-way cross-family checking and the Gini cap made this value learnable and trustworthy: on seen schemas the decisions are Bayes-optimal and the questions oracle-level, and on real conversations the questions go where answers help. The cap matters beyond our setting: an uncapped head can learn to output zero when the model is certain, and in distribution it does, but out of distribution it values fields that merely look important. Next steps are a Turkish model on MoganBERT-TR (Yilmaz et al., 2026a), more and more varied schemas to close the zero-shot gap, the ABCD training split for calibration on real dialogues, and a study with human participants.

Limitations

Synthetic dialogues: answers in the controlled study come from a simulator whose kinds and rates we chose, and LLM-written text may be more cooperative than real users; the real-data experiments test single exchanges, not the full loop with people. **Zero-shot decisions** stay 15 points below the ceiling, and **calibration out of distribution** is weak (ABCD ECE 0.19). **Residual overlap** in the privacy schema caused 22% message rejection, a mild selection effect. **Missing baselines:** a slow VOI that re-encodes every possible answer, an LLM agent, and an ablation of the self-referential target. Questions are slot **templates**; all runs use **one seed** and **English** only; Laya’s numbers are taken from its repository, and the two models are trained on different mixtures.

Ethics Statement

LAVOIR is a routing component: it chooses a unit, asks at most a few questions, or hands the case to a human. In sensitive domains (privacy, clinic routing, insurance) the hand-off threshold should be conservative and questions should be reviewed so that they do not request unnecessary personal information. All synthetic profiles are fictitious; the generator and checker (Qwen3.6-35B-A3B, Gemma-4-26B-A4B) are Apache-2.0 models that place no restriction on generated text. Most general sources are under permissive or share-alike licenses (Apache-2.0, MIT, CC0, CC BY, CC BY-SA), but several restrict use to non-commercial research: ANLI and the support tickets (CC BY-NC 4.0), MS MARCO, the Yelp reviews, MARC and QQP (their providers’ terms); the phishing corpus is LGPL-3.0, and sources without a license on the Hugging Face Hub were used under their original distribution terms. We therefore release the weights under CC BY-NC 4.0 and do not redistribute any third-party data.

Code and Data Availability

The code is available at <https://github.com/moganai/lavoir> under Apache-2.0: the model, sequence builder, losses, VOI target computation, question policy, evaluation, training and fine-tuning scripts, tests, and example workflow definitions with the data format. The final model with its fitted temperatures is available at <https://huggingface.co/moganai/lavoir> under CC BY-NC 4.0. The general single-turn mixture is not redistributed; it is built from the public sources listed in Appendix B.

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A Schemas

Gates. Every unit is reachable, unit masses lie in $[0.03, 0.5]$, at least 5% of the mass needs two or more slots, and a dry run of the exact posterior gives a mean maximum posterior in $[0.35, 0.75]$ and a share of unknown slots with $\text{VOI}^* > 0.05$ in $[0.2, 0.6]$. A validator enforces that the last rule is *else* and that non-decisive slots occur in no rule; it caught a telecom slot used in no rule, fixed by adding a field-technician unit.

Pilot history. In pilot 1 the partial-answer hint accuracy was 0.70–0.79 instead of 0.5; the bias was confined to probes of slots already stated in the message, which the checker read from context, so such probes are now always clean. Message leakage clustered on overlapping slot pairs (banking *unrecognized transaction / authorized = no*). Pilot 2 added *not applicable* values, which the generator rendered as “no” and produced 1,293–1,404 answer contradictions per schema; they were removed. Pilot 3 cut message rejection to 0.32–0.35 in banking and privacy and answer rejection to 1.6–3.6%. The first full run lost up to 36% of cases in telecom and clinic_routing because non-decisive dates implied decisive facts and the generator invented concrete problems when the topic was hidden; the final run replaced the non-decisive slot with the customer’s name and forbade invented specifics.

B Training Data

General v1 (19,000). Typed-decisions train split (6,000); CLINC150-plus with random subsets of 5–15 intents (5,000); MultiNLI entailment/contradiction as yes/no with the hypothesis as instruction (3,000); Yelp reviews as class-balanced score questions (2,000); SGD first user turns with the active intent among the intents of the service and two to four other services (3,000). Each source has 10–12 instruction wordings.

Tool choice (6,000). One of 13 shards of Nemotron-Post-Training-Dataset-v1: the state is the user message, the options are the row’s tool set with distractors, and the gold is the single

Table 4: The twelve schemas. *Profiles*: decisive-slot combinations with non-zero prior. *Multi-slot*: prior mass of units that need at least two slots. *Max post.*: mean maximum posterior of message-level evidence in the dry run. *VOI>.05*: share of unknown slots with oracle VOI above 0.05.

Schema	Split	Units	Decisive/all	Profiles	Unit mass	Multi-slot	Max post. / VOI>.05
banking_support	train	5	4/5	44	.100–.350	1.00	.474 / .498
ecommerce_returns	train	5	4/5	32	.061–.324	.70	.455 / .474
hr_requests	train	6	3/4	24	.080–.240	.80	.405 / .532
insurance_claims	train	5	4/5	32	.109–.256	1.00	.456 / .518
it_helpdesk	train	5	4/5	80	.130–.292	.70	.400 / .449
privacy_requests	train	5	4/5	76	.063–.440	1.00	.549 / .492
telecom_support	train	6	4/5	32	.090–.270	.80	.379 / .410
travel_changes	train	5	4/5	36	.053–.350	.85	.516 / .504
saas_support	zero-shot	5	4/5	44	.059–.452	.88	.562 / .468
parcel_delivery	zero-shot	5	4/5	32	.105–.341	1.00	.485 / .492
student_affairs	zero-shot	6	4/5	32	.067–.282	.80	.441 / .439
clinic_routing	zero-shot	6	4/5	32	.050–.242	.80	.400 / .431

Table 5: Per-schema leakage test on seen schemas: model minus posterior probability of the gold unit, and accuracy minus ceiling, with case-clustered bootstrap 95% intervals where the interval excludes or nearly excludes zero.

Schema	RL0		RL1	
	$p(\text{gold})$ model–post.	Acc–ceiling	$p(\text{gold})$ model–post.	Acc–ceiling
banking	−.020 [−.037, −.007]	−.054 [−.100, −.013]	−.014 [−.027, −.004]	−.038 [−.074, −.002]
ecommerce	+.001 [−.002, .003]	−.016	−.000	−.010
hr	−.002 [−.005, .000]	−.000	+.003 [.000, .005]	+.003
insurance	−.005 [−.008, −.002]	+.041 [.002, .080]	−.000 [−.005, .004]	+.038 [−.001, .076]
it_helpdesk	−.003	−.006	+.002	−.008
privacy	−.009 [−.015, −.005]	+.021 [−.015, .056]	−.004	+.018
telecom	−.004	−.025	−.001	−.034 [−.067, −.001]
travel	−.002	−.016	−.003	−.017

tool called in the first assistant turn. We removed rows whose gold tool name appears in the user message (2,930, a dataset artifact), rows without a call or with several distinct calls, and set sizes outside 2–12; train and test (5,447 / 553) are split by tool set. LLM-written tool schemas were explored as VOI schemas but dropped: only 2 of 200 passed the gates, and an assistant does not ask the user which tool to call.

Laya sources (19,437). AG News train (2,500), BoolQ (3,000), Enron spam train (1,937), phishing e-mails (2,000), MS MARCO v1.1 train (2,500) and support tickets (2,500), plus question-form yes/no items from CLINC, Yelp, MultiNLI and SGD (5,000). All 4,447 state texts of Laya’s benchmark builder and 50 further Enron subject-line overlaps were removed.

Open diversity (38,200, 27 sources, train splits only). Choice (17,000): DBpedia and Yahoo Answers (Zhang et al., 2015), 20 Newsgroups (Lang, 1995), Tweet Topic (Antypas et al., 2022), GoEmotions (Demszky et al., 2020), TweetEval emotion and sentiment (Barbieri et al., 2020), MATH topic (Hendrycks et al., 2021), Dolly task type (Conover et al., 2023), and domain routing among GSM8K, MATH, MBPP, Writing-Prompts, Natural Questions, SQL questions and Dolly (Cobbe

et al., 2021; Austin et al., 2021; Fan et al., 2018; Kwiatkowski et al., 2019; b-mc2, 2023). Yes/no (17,400): SST-2 (Socher et al., 2013), IMDB (Maas et al., 2011), Civil Comments (Borkan et al., 2019), Measuring Hate Speech (Kennedy et al., 2020), TweetEval hate, offensive and irony, QQP (Iyer et al., 2017), PAWS (Zhang et al., 2019), MRPC (Dolan and Brockett, 2005), SNLI (Bowman et al., 2015), ANLI (Nie et al., 2020), SciTail (Khot et al., 2018), CoLA (Warstadt et al., 2019). Score (3,800): Amazon star ratings from the English part of MARC (Keung et al., 2020) (SetFit copy), toxicity level from the Civil Comments toxicity score, and hate level from the Measuring Hate Speech score. Multi-class sources use random option subsets, 30% of negatively phrased questions have inverted targets, and states are often dictionaries with named fields. None of Laya’s evaluation sets was used.

C Additional Results

Leakage. Table 5: for insurance the model assigns the gold unit slightly *less* probability than the posterior; the argmax excess comes from nearly tied posteriors (e.g. 0.44/0.56), and banking shows the same noise in the opposite direction. The examples on which the model is most confident relative to the posterior contain no hidden clue, only partial answers

ambiguous between two values.

Policy gradient. Table 6 shows the miniature model and Table 7 the full-scale controlled chains. At full scale the ratio between the policy-gradient and VOI gradient norms on the shared parameters was $1.5\text{--}7\times$.

Table 6: Miniature model ($d = 128$, 100 examples, $w_{\text{RL}} = 1$): decision KL, VOI loss and gradient norms sent to the shared parameters.

Step	KL	VOI MSE	$\ g_{\text{RL}}\ $	$\ g_{\text{CE}}\ $	$\ g_{\text{VOI}}\ $
0	.511	1.730	0.017	0.001	11.50
100	.033	0.984	48.49	0.491	0.225
300	.011	0.673	28.71	0.158	8.384
600	.007	0.127	26.95	0.098	2.445

Table 7: Full-scale controlled training on the calibration slice: warm phase at $T = 1$, and joint phase (VOI Spearman and MSE against sampled targets; KL on our schemas at $T = 1$ and after fitting one temperature per type).

	$w_{\text{RL}} = 1$	$w_{\text{RL}} = 0$
warm ep. 1: acc / KL	.807 / .230	.838 / .102
warm ep. 2: acc / KL	.857 / .073	.896 / .018
warm ep. 3: acc / KL	.896 / .019	.927 / .006
joint: Spearman / MSE	.489 / .0108	.527 / .0104
joint: acc all / ours	.919 / .965	.918 / .956
joint: KL, $T=1 \rightarrow$ fitted	.0061 $\rightarrow$.0612	.0040 $\rightarrow$.0381

Temperatures. Per-group temperatures give $T = 1.015\text{--}1.017$ for our schemas and $T = 3.78\text{--}4.34$ (choice), $5.52\text{--}7.51$ (score) and $2.04\text{--}4.69$ (yes/no) for the general data, whose KL drops from $0.70\text{--}0.94$ to $0.28\text{--}0.31$.

Effect of the broad data. Broadening the data kept the dialogue behaviour on seen schemas (AUC $0.799 \rightarrow 0.795$, oracle 0.797 ; accuracy $0.652 \rightarrow 0.648$ against a ceiling of 0.659) but lowered VOI rank correlation ($0.831 \rightarrow 0.763$) and zero-shot b0.5 ($0.579 \rightarrow 0.535$), which the Gini cap restored to 0.581 . Shuffling the options changes 1% of AG News, 5% of Emotion, 12% of MASSIVE and 29% of Banking77 decisions of the final model.

D Implementation

The experiments used an internal module on top of Laya’s code (version 0.3.11, commit 1e28ac2) that leaves Laya’s files unchanged; the released `lavo` package re-implements it as a standalone package with its own tests. The internal module’s 149 tests cover bit-identical logits with no slots, marker positions under shuffling, the posterior against brute-force enumeration, zero VOI for known and non-decisive slots, $0 \leq \hat{v}_k \leq G(p)$ under the cap, no VOI gradient in the option

scorer, and the policy. Laya’s act/escalate head is trained with a zero-weighted loss in its notebook, so we do not use it.